

Douglas J. Williams

Technical Product Owner | AI-Assisted Software Delivery

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Open to fully remote opportunities

PROFESSIONAL SUMMARY

Technical Product Owner and AI-assisted software builder with 28+ years in mission-critical network operations, product leadership, and internal-tool delivery. I define product vision, translate business needs into requirements, and direct AI-assisted development from specification through testing and debugging to a working result. At CPKC, I drove a company-wide event-analysis platform that processes billions of PTC messages per day for proactive problem isolation, and independently built and run a production app tracking 58 oil & gas wells across 6 operators. I pair deep systems knowledge with practical delivery — directing AI to build in Python, SQL, React, and Supabase. Seeking a fully remote Technical Product Owner or AI Product Owner role focused on internal tools, operational applications, and business-driven AI solutions.

CORE COMPETENCIES

Product Ownership: Product vision & requirements definition · business-to-technical translation · scope & delivery ownership · operational validation · stakeholder & vendor coordination

AI-Assisted Development: Spec-driven development · directing AI coding agents (Kiro) · requirements-driven delivery · iterative testing & debugging to a working result

Software & Data (AI-directed): Python · React · Supabase/Postgres · SQL · Oracle · data ingestion · Git/GitHub · Vercel · HTML/CSS

Technical Operations: Kubernetes · Linux/command line · log analysis at scale · incident response · 24/7 mission-critical support · technical documentation (MOPs)

Domain Expertise: Positive Train Control (PTC) · telecom & network operations · SMS/MMS platforms · IMS Core · oil & gas

PROFESSIONAL EXPERIENCE

Senior PTC Applications Engineer — Canadian Pacific Kansas City (CPKC)

2022 – Present · Remote

- Own the vision and delivery of internal tools supporting Positive Train Control operations across a safety-critical rail network.
- Drove development of a company-wide event-analysis platform processing billions of PTC messages per day for detailed event analysis and proactive problem isolation.
- Delivered a platform used across CPKC to surface operational issues before they impact operations.
- Delivered Python database-integration tools connecting SQL and Oracle data sources into custom applications and reporting workflows.
- Define requirements and target outcomes for AI-assisted, spec-driven development; direct implementation and perform iterative testing and debugging through to a working result.
- Apply deep operational knowledge to validate software behavior against real-world PTC requirements.

ATS Senior Network Engineer — T-Mobile

Aug 2020 – Aug 2022

- Product Lead for MCO (Message Controller) and Kubernetes-containerized MCO handling SMS/MMS traffic.
- Managed and implemented migration of Sprint traffic to a new Replica MCO environment, enabling contract expirations and consolidation onto a single network.

- Led design and implementation efforts; drove proactive tooling and processes to identify and resolve issues before customer impact.
- Authored Methods of Procedure (MOPs) and trained teams on new products during rollout.

Network Translations Specialist / NTAC Network Engineer — Sprint

Jul 1998 – Aug 2020

- Implemented a virtual SMS/MMS messaging platform with Mavenir and Rackspace; led SMS migration to the IMS Core via SIP with CDMA fallback.
- Migrated MMS storage and processing from Syniverse to an internal virtualized MMSC, eliminating outside managed-services costs.
- Saved Sprint **\$1.2M** over three years by renegotiating the SPAM-filtering vendor contract; provided 24/7/365 support during high-severity events.

SELECTED PRODUCT & SOFTWARE PROJECTS

PTC Event-Analysis Platform — Python · SQL/Oracle · AI-assisted (CPKC)

Company-wide operational platform processing billions of PTC messages per day. Owned the vision and requirements, drove development, and validated the result through testing and debugging.

Oil & Gas Well-Tracker Application — React · Supabase/Postgres · Python · Vercel (J&T Cook Corporation, own business)

Designed and delivered a full-stack app to track working-interest investments — investment vs. revenue, payout status, operator-statement ingestion, bank reconciliation, and tax reporting. Tracks 58 working-interest wells (48 currently producing) across 6 operators, ingesting monthly operator settlement statements and reconciling roughly 170 net barrels of oil per month against bank activity. Owned the product vision and requirements; directed AI-assisted implementation and debugged through to live production deployment. Sanitized demo:

github.com/djwilli69/well-tracker-demo

Database Integration Tools — Python · SQL · Oracle · AI-assisted (CPKC)

Utilities connecting operational database sources into custom applications and reporting workflows; defined requirements, directed implementation, and tested against real operational needs.

BUSINESS OWNERSHIP

Owner — J&T Cook Corporation (Oil & Gas)

Own and operate an oil & gas business with working interests in 58 wells across 6 operators, including building its software with AI-assisted development. Bring a product owner's mindset: I understand the business goal behind the requirements and deliver tools that support operational and financial needs.

EDUCATION

Bachelor of Science, Business Management — Friends University, Wichita, KS